Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If elements with principal quantum number $n > 4$ were not allowed in nature, the number of possible elements would be

(A) 60

(B) 32

(C) 4

(D) 64

Q.2

An α -particle of energy $5\mathrm{MeV}$ is scattered through 180° by a fixed uranium nucleus. The distance of closest approach is of the order of

(A) 10^{-12} cm

(B) 10^{-10} cm

(C) 10^{-14} cm

(D) 10^{-15} cm

Q.3

In a Rutherford experiment, the number of particles scattered at 90° angle are 28 per minute then number of scattered particles at an angle 60° and 120° will be

(A) 117 per minute, 25 per minute

(B) 50 per minute, 12.5 per minute

(C) 100 per minute, 200 per minute

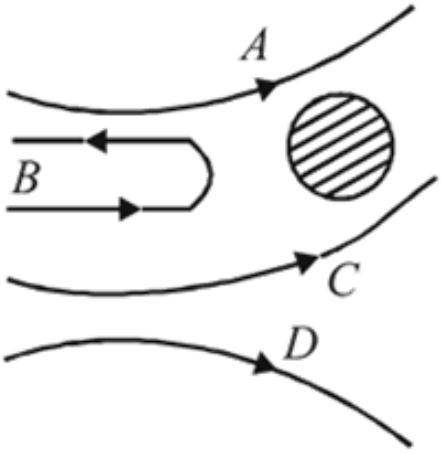
(D) 112 per minute, 12.4 per minute

Q.4

In the Rutherford experiment, α -particles are scattered from a nucleus as shown. Out of the four paths,

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which path is not possible?



(A) D

(B) B

(C) C

(D) A

Q.5

In Rutherford scattering experiment, what will be the correct angle for α -scattering for an impact parameter, $b = 0$?

(A) 90°

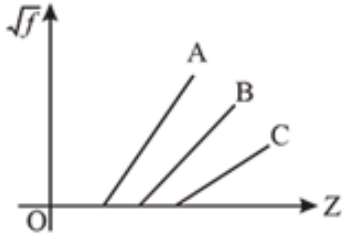
(B) 270°

(C) 0°

(D) 180°

Q.6

Figure shows Moseley's plot between $\sqrt{f}$ and Z, where f is the frequency and Z is the atomic number. Three lines A, B and C shown in the graph may represent



(A) $\mathrm{K}_\alpha, \mathrm{K}_\beta$, and K_γ lines, respectively

(B) $\mathrm{K}_\gamma, \mathrm{K}_\beta$, and K_α lines, respectively

(C) $\mathrm{K}_\alpha, \mathrm{L}_a$, and M_α lines, respectively

(D) None of these

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Q.7

The energy of He^{+} in the ground state is $-54.4\ eV$, then the energy of Li^{++} in the first excited state will be

(A) $-30.6eV$

(B) $27.2eV$

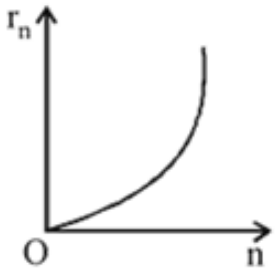
(C) $-13.6eV$

(D) $-27.2eV$

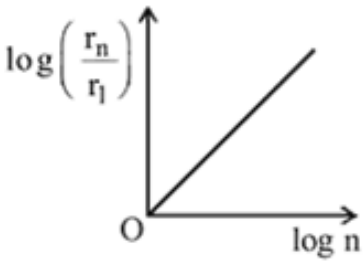
Q.8

If in hydrogen atom, radius of nth Bohr orbit is r_n, frequency of revolution of electron in n^{th} orbit is f_n, choose the correct option.

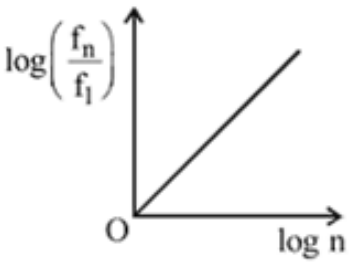
(A)



(B)



(C)



(D) Both (a) and (b)

Q.9

If one were to apply Bohr model to a particle of mass ' m ' and charge ' q ' moving in a plane under the influence of a magnetic field 'B', the energy of the charged particle in the nth level will be:

(A) $n\left(\frac{hqB}{2\pi m}\right)$

(B) $n\left(\frac{hqB}{8\pi m}\right)$

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(C) $\mathrm{n}\left(\frac{\mathrm{hqB}}{4\pi \mathrm{m}}\right)$

(D) $\mathrm{n}\left(\frac{\mathrm{hqB}}{\pi \mathrm{m}}\right)$

Q.10

The acceleration of an electron in the first orbit of the hydrogen atom $(\mathrm{z}=1)$ is:

(A) $\frac{h^2}{\pi^2 m^2 r^3}$

(B) $\frac{h^2}{8\pi^2 m^2 r^3}$

(C) $\frac{h^2}{4\pi^2 m^2 r^3}$

(D) $\frac{h^2}{4\pi m^2 r^3}$

Q.11

An excited state of doubly ionised Lithium $\left(\mathrm{Li}^{2+}\right)$ has an orbital radius that is about 1.33 times that of the ground state of hydrogen (H) (in Bohr's theory). The ratio of energy of the two states, $E\left(\mathrm{Li}^{2+}\right) / E(\mathrm{H})$, is

(A) 2.25

(B) 4.5

(C) 1

(D) 9

Q.12

In the Bhor s model of hydrogen-like atom the force between the nucleus and the electron is modified as $F=\frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{r^2}+\frac{\beta}{r^3}\right)$, where β is a constant. For this atom, the radius of the nth orbit in terms of the Bohr radius $\left(a_0=\frac{\varepsilon_0 h^2}{m\pi e^2}\right)$ is:

(A) $r_n=a_0 n-\beta$

(B) $r_n=a_0 n^2+\beta$

(C) $r_n=a_0 n^2-\beta$

(D) $r_n=a_0 n+\beta$

Q.13

In the Bohr model of a hydrogen atom, the centripetal force is furnished by the coulomb attraction

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between the proton and the electron. If a_0 is the radius of the ground state orbit, m is the mass, e is the charge on the electron and ε_0 is the vacuum permittivity, the speed of the electron is

(A) 0

(B) $\frac{e}{\sqrt{\varepsilon_0 a_0 m}}$

(C) $\frac{e}{\sqrt{4\pi\varepsilon_0 a_0 m}}$

(D) $\frac{\sqrt{4\pi\varepsilon_0 a_0 m}}{e}$

Q.14

A hydrogen atom and a doubly ionized lithium atom are both in the second excited state. If L_H and L_{Li} respectively represent their electronic angular momentum and E_H and E_{Li} their energies, then

(A) $L_H > L_{Li}$ and $|E_H| > |E_{Li}|$

(B) $L_H = L_{Li}$ and $|E_H| < |E_{Li}|$

(C) $L_H = L_{Li}$ and $|E_H| > |E_{Li}|$

(D) $L_H < L_{Li}$ and $|E_H| < |E_{Li}|$

Q.15

As per Bohr model, the minimum energy (in eV) required to remove an electron from the ground state of doubly ionized Li atom $(Z = 3)$ is

(A) 1.51

(B) 13.6

(C) 40.8

(D) 122.4

Q.16

Ionization energy of a hydrogen-like ion A is greater than that of another hydrogen-like ion B. If $\mathrm{r}, \mathrm{u}, \mathrm{E}$ and L represent the radius of the orbit, speed of the electron, energy of the atom and orbital angular momentum of the electron respectively then in ground state

(A) $\mathrm{r_A} > \mathrm{r_B}$

(B) $\mathrm{u_A} > \mathrm{u_B}$

(C) $\mathrm{E_A} > \mathrm{E_B}$

(D) $\mathrm{L_A} > \mathrm{L_B}$

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Q.17

An electron in a hydrogen atom makes a transition from first excited state to ground state. The equivalent current due to circulating electron-

(A) increases 2 times

(B) increases 4 times

(C) increases 8 times

(D) remains the same

Q.18

According to Bohr's theory, the time averaged magnetic field at the centre (i.e. nucleus) of a hydrogen atom due to the motion of electrons in the nth orbit is proportional to : $(n =$ principal quantum number)

(A) n^{-4}

(B) n^{-5}

(C) n^{-3}

(D) n^{-2}

Q.19

Find the inverse ratio of maximum to minimum wavenumber in Lyman series.

(A) $9:5$

(B) $4:3$

(C) $36:5$

(D) $4:1$

Q.20

K_α wavelength emitted by an atom of atomic number $Z = 11$ is λ. Find the atomic number for an atom that emits K_α radiation with wavelength 4λ.

(A) 6

(B) 5

(C) 7

(D) 4

Answers & Solutions

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Q.1 Answer:

60

Solution:

The maximum number of electrons in an orbit is $2n^2$. $n > 4$ is not allowed. Therefore the number of maximum electron that can be in first four orbits are

$2(1)^2 + 2(2)^2 + 2(3)^2 + 2(4)^2$

$= 2 + 8 + 18 + 32 = 60$

Therefore, possible element are 60 .

Q.2 Answer:

10^{-12} cm

Solution:

Distance of closest approach

$r_0 = \frac{Ze(2e)}{4\pi\varepsilon_0\left(\frac{1}{2}mv^2\right)}$

Energy, $E = 5 \times 10^6 \times 1.6 \times 10^{-19}$ J

$\therefore r_0 = \frac{9\times10^9\times\left(92\times1.6\times10^{-19}\right)\left(2\times1.6\times10^{-19}\right)}{5\times10^6\times1.6\times10^{-19}}$

$\Rightarrow r = 5.2 \times 10^{-14}\ \text{m} = 5.3 \times 10^{-12}\ \text{cm}$

Q.3 Answer:

112 per minute, 12.4 per minute

Solution:

No. of particles scattered through an angle $\theta = \mathrm{N}(\theta) = \frac{\mathrm{k}Z^2}{\sin^4\left(\frac{\theta}{2}\right)(\mathrm{K.E.})^2}$

$\therefore 28 = \frac{4\mathrm{kcz}^2}{(\mathrm{K.E.})^2}$ **for** $\theta = 90^\circ$

$\therefore \frac{\mathrm{kz}^2}{(\mathrm{K.E.})^2} = \frac{28}{4} = 7$

$\therefore \mathrm{N}\,(60^\circ) = \frac{7}{\sin^4\left(\frac{60^\circ}{2}\right)} = 16 \times 7 = 112/\mathrm{min.}$

$\mathrm{N}\,(120^\circ) = \frac{7}{\sin^4\left(\frac{120^\circ}{2}\right)} = 12.4/\mathrm{min}$

Q.4 Answer:

C

Solution:

As α **-particles are doubly ionised helium** He^{++} **i.e. Positively charged and nucleus is also positively charged and we know that like charges repel each other.**

Q.5 Answer:

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180°

Solution:

$b = \frac{Ze^2 \cot\left(\frac{\theta}{2}\right)}{4\pi\epsilon_0 k_i} = 0 \Rightarrow \cot\left(\frac{\theta}{2}\right) = 0$
$\Rightarrow \frac{\theta}{2} = 90^\circ$ **or** $\theta = 180^\circ$

Q.6 Answer:

None of these

Solution:

The K, L and M lines have different intercepts. The intercept of K is more than that of L, which in turn is more than that of M.

Q.7 Answer:

-30.6eV

Solution:

Energy of electron in n^{th} orbit is $E_n = -(Rch)\frac{Z^2}{n^2} = -54.4\text{eV}$
For He^+ is ground state
$E_1 = -(Rch)\frac{(2)^2}{(1)^2} = -54.4 \Rightarrow Rch = 13.6$
$\therefore$ **For Li^{++} in first excited state** $(n = 2)$
$E' = -13.6 \times \frac{(3)^2}{(2)^2} = -30.6\text{eV}$

Q.8 Answer:

Both (a) and (b)

Solution:

Radius of nth orbit $r_n \propto n^2$, graph between r_n and n is a parabola. Also, $\frac{r_n}{r_1} = \left(\frac{n}{1}\right)^2 \Rightarrow \log_e\left(\frac{r_n}{r_1}\right) = 2\log_e(n)$
Comparing this equation with $y = mx + c$, Graph between $\log_e\left(\frac{r_n}{r_1}\right)$ and $\log_e$ (n) will be a straight line, passing from origin. Similarly it can be proved that graph between $\log_e\left(\frac{f_n}{f_1}\right)$ and $\log_e n$ is a straight line.
But with negative slops.

Q.9 Answer:

$n\left(\frac{hqB}{4\pi m}\right)$

Solution:

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$qVB = \frac{mv^2}{r}$ **and** $\frac{nh}{2\pi} = mvr$

Multiplying equation (i) and (ii),

$\frac{qBnh}{2\pi} = m^2v^2$

Now multiplying both sides by $\frac{1}{2m}$

$n\frac{qBh}{4\pi m} = \frac{1}{2}mv^2 \quad$ **i.e.** $\mathrm{KE} = n\left[\frac{qBh}{4\pi m}\right]$

Q.10 Answer:

$\frac{h^2}{4\pi^2 m^2 r^3}$

Solution:

Speed of electron in first orbit $(n = 1)$ **of hydrogen atom** $(\mathrm{z} = 1)$**,**

$\mathrm{v} = \frac{\mathrm{e}^2}{2\varepsilon_0\,\mathrm{h}} \quad \mathrm{r} = \frac{\mathrm{h}^2\varepsilon_0}{\pi \mathrm{me}^2} \Rightarrow \varepsilon_0 = \frac{\mathrm{r}\pi \mathrm{me}^2}{\mathrm{h}^2}$

Acceleration of electron, $\frac{v^2}{r} = \frac{e^4}{4\varepsilon_0^2 h^2} \times \frac{\pi m e^2}{h^2 \varepsilon_0}$

$= \frac{\mathrm{e}^4 \times \pi \mathrm{me}^2}{4\,\mathrm{h}^4 \varepsilon_0^3}$ **....(ii)**

eliminating ε_0

$= \frac{e^4 \pi m e^2 h^6}{4h^4 r^3 \pi^3 m^3 e^6} = \frac{h^2}{4\pi^2 m^2 r^3}$

Q.11 Answer:

9

Solution:

Considering it to be doubly ionised

$\frac{r_{\mathrm{Li}^{+2}}}{r_H} = \frac{n_{L_i i^2}}{n_H} \times \frac{Z_H}{Z_{L^{+2}}}$

$1.33 = \frac{n_{L^{+2}}}{1} \times \frac{1}{3}$

$n_{\mathrm{Li}^{+2}} = 3.99 \approx 4$

$E_{\mathrm{Li}^{-12}} = -13.6\frac{Z^2}{n^2} \Rightarrow -13.6 \times \frac{3^2}{1^2}$

$E_H = -13.6 \times \frac{Z^2}{n^2} \Rightarrow -13.6$

$E_{L^{+2}} : E_H = 9 : 1$

Q.12 Answer:

$r_n = a_0 n^2 - \beta$

Solution:

As $\mathrm{F} = \frac{\mathrm{mv}^2}{\mathrm{r}} = \frac{\mathrm{e}^2}{4\pi \in_0}\left(\frac{1}{\mathrm{r}^2} + \frac{\beta}{\mathrm{r}^3}\right)$

and $\mathrm{mvr} = \frac{\mathrm{nh}}{2\pi} \Rightarrow \mathrm{v} = \frac{\mathrm{nh}}{2\pi \mathrm{mr}}$

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$\therefore \mathrm{m}\left(\frac{\mathrm{nh}}{2\pi \mathrm{mr}}\right)^2 \times \frac{1}{\mathrm{r}}=\frac{\mathrm{e}^2}{4\pi\epsilon_0}\left(\frac{1}{\mathrm{r}^2}+\frac{\beta}{\mathrm{r}^3}\right)$

or, $\frac{1}{\mathrm{r}^2}+\frac{\beta}{\mathrm{r}^3}=\frac{\mathrm{mn}^2\,\mathrm{h}^2 4\pi\in_0}{4\pi^2\,\mathrm{m}^2\mathrm{e}^2\mathrm{r}^3}$

or, $\frac{\mathrm{a_0 n}^2}{\mathrm{r}^3}=\frac{1}{\mathrm{r}^2}+\frac{\beta}{\mathrm{r}^3}$ **(**$\because \mathrm{a_0}=\frac{\epsilon_0\,\mathrm{h}^2}{\mathrm{m}\pi\mathrm{e}^2}$ **Given)**

$\therefore \mathrm{r}=\mathrm{a_0 n}^2-\beta$

Q.13 Answer:

$\frac{e}{\sqrt{4\pi\varepsilon_0 a_0 m}}$

Solution:

Centripetal force = Coulombian force

$\frac{mv^2}{a_0}=\frac{1}{4\pi\varepsilon_0}\cdot\frac{e\times e}{a_0^2}$

$\Rightarrow v^2=\frac{e^2}{4\pi\varepsilon_0\cdot a_0\cdot m}\Rightarrow v=\frac{e}{\sqrt{4\pi\varepsilon_0\cdot a_0\cdot m}}$

Q.14 Answer:

$L_H = L_{Li}$ and $|E_H| < |E_{Li}|$

Solution:

fiven $Z_{\mathrm{H}}=1,\quad \mathrm{Z_{La}}=3$ **and** $\mathrm{nn}=\mathrm{r_L}=3$

Angular momentum is independent of z**, therefore** $l_t = 1_{Li}$

Energy is directiy proportional to square of atomic number, hence $\mathrm{E_H} < \mathrm{E_{La}}$

Q.15 Answer:

122.4

Solution:

$E_n=-13.6\frac{\left(Z^2\right)}{\left(n^2\right)}\mathrm{eV}$

Therefore, ground state energy of doubly ionized lithium $\mathrm{atom}(Z=3, n=1)$ **will be**

$E_1=(-13.6)\frac{(3)^2}{(1)^2}=-122.4\mathrm{eV}$

∴ Ionization energy of an electron in ground state of doubly ionized lithium atom will be

$122.4\mathrm{eV}$

Q.16 Answer:

$\mathrm{u_A} > \mathrm{u_B}$

Solution:

$\varepsilon_A > \varepsilon_B$

$\mu_A > \mu_B$

Q.17 Answer:

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increases 8 times

Solution:

$i = \frac{q}{T}$ **Now,** $T^2 \propto r^3 \propto n^6 \Rightarrow i \propto \frac{1}{n^3}$

Q.18 Answer:

n^{-5}

Solution:

$$B = \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I_n}{2r_n}$$

$$r_n = \frac{n^2 h^2}{4r^2 mke^2}$$

$$\Rightarrow r_n \propto n^2$$

$$f = \frac{v}{2rr} = \frac{2rke^2}{nh \cdot 2rr}$$

$$\Rightarrow f_n \propto \frac{1}{nr_n} \Rightarrow f_n \alpha \frac{1}{n^3}$$

$$I = q/t = qf^2$$

$$\Rightarrow I \propto \frac{1}{n^3}$$

$$\therefore B \propto \frac{I}{r} = \frac{n^{-3}}{n^2} = n^{-5}$$

Q.19 Answer:

4: 3

Solution:

For maximum wavelength λ_{max}

$n_1 = 1, n_2 = 2$

$\frac{1}{\lambda_{max}} = R\frac{1}{(1)^2} - \frac{1}{(2)^2}$

$\lambda_{max} = \frac{4}{3\pi}$

For minimum wavelength λ_{min}

$n_1 = 1, n_2 = \infty$

$\frac{1}{\lambda_{min}} = R$

$\lambda_{min} = \frac{1}{R}$

$\frac{\lambda_{max}}{\lambda_{min}} = \frac{4}{3R\frac{1}{R}} = \frac{4}{3}$

Q.20 Answer:

6

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Solution:

$\frac{1}{\lambda} \propto z\text{-}1^2$

$\therefore \frac{\lambda_1}{\lambda_2} = \frac{z_2 - 1^2}{z_1 - 1^2}$

$\frac{1}{4} = \frac{Z_2 - 1^2}{11 - 1}$

Solving this, we get, $Z_2 = 6$

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